MANAJEMEN BASIS DATA

Stored Procedure, SQL Constraints, dan Otomatisasi Database

Dosen Pengampu:

Satriyo Adhy, S.Si., M.T

DISUSUN OLEH

Muhammad Fikri	24060124130069
Nayla Husna	24060124140158
Muhammad Akmal Fazli Riyadi	24060124130123
Muhammad Fahri	24060124120037
Khanza Qaila	24060124120011

PROGRAM STUDI S1-INFORMATIKA

UNIVERSITAS DIPONEGORO

SEMARANG

2026

PEMBAHASAN

DDL DAN DML

```
SET FOREIGN_KEY_CHECKS = 0;

TRUNCATE TABLE tiket;
TRUNCATE TABLE jadwal;
TRUNCATE TABLE stasiun;
TRUNCATE TABLE kereta;
TRUNCATE TABLE penumpang;

SET FOREIGN_KEY_CHECKS = 1;


-- 1. TABEL KERETA
CREATE TABLE kereta (
    id_kereta VARCHAR(5) PRIMARY KEY,
    nama_kereta VARCHAR(50) NOT NULL,
    kapasitas INT NOT NULL
);

-- 2. TABEL STASIUN
CREATE TABLE stasiun (
    id_stasiun INT PRIMARY KEY AUTO_INCREMENT,
    kode_stasiun VARCHAR(10) UNIQUE NOT NULL,
    nama_stasiun VARCHAR(100) NOT NULL,
    kota VARCHAR(50) NOT NULL,
    wilayah_daop VARCHAR(20)
);

-- 3. TABEL PENUMPANG
CREATE TABLE penumpang (
    id_penumpang INT PRIMARY KEY AUTO_INCREMENT,
    nik VARCHAR(20) NOT NULL UNIQUE,
    nama_lengkap VARCHAR(100) NOT NULL,
    jenis_kelamin ENUM('L', 'P') NOT NULL,
    no_telepon VARCHAR(15),
    email VARCHAR(30),
    tanggal_daftar DATE
);

-- 4. TABEL JADWAL
CREATE TABLE jadwal (
    id_jadwal INT AUTO_INCREMENT PRIMARY KEY,
    id_kereta VARCHAR(5) NOT NULL,
    id_stasiun_asal INT NOT NULL,
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    id_stasiun_tujuan INT NOT NULL,
    waktu_berangkat DATETIME NOT NULL,
    waktu_tiba DATETIME NOT NULL,
    harga INT NOT NULL,

    FOREIGN KEY (id_kereta) REFERENCES kereta(id_kereta),
    FOREIGN KEY (id_stasiun_asal) REFERENCES stasiun(id_stasiun),
    FOREIGN KEY (id_stasiun_tujuan) REFERENCES
stasiun(id_stasiun)
);

-- 5. TABEL TIKET
CREATE TABLE tiket (
    id_tiket INT PRIMARY KEY AUTO_INCREMENT,
    kode_booking VARCHAR(20) NOT NULL UNIQUE,
    id_penumpang INT NOT NULL,
    id_jadwal INT NOT NULL,
    nomor_gerbong INT NOT NULL,
    nomor_kursi VARCHAR(5) NOT NULL,
    waktu_pemesanan DATETIME NOT NULL,
    status_pembayaran VARCHAR(20) NOT NULL,
    metode_pembayaran VARCHAR(30) NOT NULL,

    FOREIGN KEY (id_penumpang) REFERENCES
penumpang(id_penumpang),
    FOREIGN KEY (id_jadwal) REFERENCES jadwal(id_jadwal)
);



INSERT INTO penumpang (nik, nama_lengkap, jenis_kelamin,
no_telepon, email, tanggal_daftar) VALUES
('3171010101900001', 'Budi Santoso', 'L', '081234567001',
'budi.santoso@gmail.com', '2023-10-01'),
('3273020202910002', 'Siti Aminah', 'P', '085678900002',
'siti.aminah@yahoo.com', '2023-10-02'),
('3374030303920003', 'Agus Setiawan', 'L', '081345678003',
'agus.setia@gmail.com', '2023-10-03'),
('3471040404930004', 'Ayu Lestari', 'P', '089654321004',
'ayu.lestari@outlook.com', '2023-10-04'),
('3578050505940005', 'Wahyu Hidayat', 'L', '081122334005',
'wahyuhidayat@gmail.com', '2023-10-05'),
('3172060606950006', 'Sri Rahayu', 'P', '082233445006',
'sri.rahayu95@gmail.com', '2023-10-06'),
('3274070707960007', 'Hendra Gunawan', 'L', '081566778007',
'hendragunawan@yahoo.com', '2023-10-07'),
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('3375080808970008', 'Dewi Sartika', 'P', '087788990008',
'dewi.sartika@gmail.com', '2023-10-08'),
('3472090909980009', 'Eko Saputra', 'L', '081900112009',
'eko.saputra@outlook.com', '2023-10-09'),
('3579101010990010', 'Rini Susanti', 'P', '083811223010',
'rini.susanti@gmail.com', '2023-10-10'),
('3173111111900011', 'Dwi Yulianto', 'L', '085722334011',
'dwi.yuli@yahoo.com', '2023-10-11'),
('3275121212910012', 'Nur Hidayah', 'P', '081233445012',
'nur.hidayah@gmail.com', '2023-10-12'),
('3376131313920013', 'Tri Wibowo', 'L', '089544556013',
'tri.wibowo@gmail.com', '2023-10-13'),
('3473141414930014', 'Sari Indah', 'P', '081355667014',
'sari.indah@yahoo.com', '2023-10-14'),
('3580151515940015', 'Anton Syahputra', 'L', '082166778015',
'anton.syah@gmail.com', '2023-10-15'),
('3174161616950016', 'Ratna Kumala', 'P', '087877889016',
'ratna.kumala@outlook.com', '2023-10-16'),
('3276171717960017', 'Yudi Pratama', 'L', '081588990017',
'yudi.pratama@gmail.com', '2023-10-17'),
('3377181818970018', 'Fitriani', 'P', '081999001018',
'fitriani.97@gmail.com', '2023-10-18'),
('3474191919980019', 'Iwan Fals', 'L', '083800112019',
'iwan.f@yahoo.com', '2023-10-19'),
('3581202020990020', 'Putri Maharani', 'P', '085711223020',
'putri.maharani@gmail.com', '2023-10-20'),
('3175212121900021', 'Arif Rahman', 'L', '081222334021',
'arif.rahman@gmail.com', '2023-10-21'),
('3277222222910022', 'Rina Marlina', 'P', '089633445022',
'rina.marlina@yahoo.com', '2023-10-22'),
('3378232323920023', 'Joko Susilo', 'L', '081344556023',
'joko.susilo@outlook.com', '2023-10-23'),
('3475242424930024', 'Eka Fitri', 'P', '082155667024',
'eka.fitri@gmail.com', '2023-10-24'),
('3582252525940025', 'Rizal Fahmi', 'L', '087866778025',
'rizal.fahmi@gmail.com', '2023-10-25'),
('3176262626950026', 'Maya Sari', 'P', '081577889026',
'maya.sari@yahoo.com', '2023-10-26'),
('3278272727960027', 'Rahmat Hidayat', 'L', '081988990027',
'rahmat.h@gmail.com', '2023-10-27'),
('3379282828970028', 'Nita Amelia', 'P', '083899001028',
'nita.amelia@gmail.com', '2023-10-28'),
('3476292929980029', 'Aditya Nugraha', 'L', '085700112029',
'aditya.nugraha@outlook.com', '2023-10-29'),
('3583303030990030', 'Wulan Guritno', 'P', '081211223030',
'wulan.g@gmail.com', '2023-10-30'),
```

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('3177313131900031', 'Bayu Pradana', 'L', '089622334031',
'bayu.pradana@yahoo.com', '2023-11-01'),
('3279323232910032', 'Yuni Shara', 'P', '081333445032',
'yuni.shara@gmail.com', '2023-11-02'),
('3380333333920033', 'Dimas Anggara', 'L', '082144556033',
'dimas.anggara@gmail.com', '2023-11-03'),
('3477343434930034', 'Anita Rachman', 'P', '087855667034',
'anita.rachman@outlook.com', '2023-11-04'),
('3584353535940035', 'Fadli Zon', 'L', '081566778035',
'fadli.zon@gmail.com', '2023-11-05'),
('3178363636950036', 'Bella Saphira', 'P', '081977889036',
'bella.saphira@yahoo.com', '2023-11-06'),
('3280373737960037', 'Gilang Dirga', 'L', '083888990037',
'gilang.dirga@gmail.com', '2023-11-07'),
('3381383838970038', 'Citra Kirana', 'P', '085799001038',
'citra.kirana@gmail.com', '2023-11-08'),
('3478393939980039', 'Hafiz Suwandi', 'L', '081200112039',
'hafiz.suwandi@outlook.com', '2023-11-09'),
('3585404040990040', 'Dian Sastrowardoyo', 'P', '089611223040',
'dian.sastro@gmail.com', '2023-11-10'),
('3179414141900041', 'Iqbal Ramadhan', 'L', '081322334041',
'iqbal.ramadhan@yahoo.com', '2023-11-11'),
('3281424242910042', 'Elita Sari', 'P', '082133445042',
'elita.sari@gmail.com', '2023-11-12'),
('3382434343920043', 'Kevin Julio', 'L', '087844556043',
'kevin.julio@gmail.com', '2023-11-13'),
('3479444444930044', 'Farah Quinn', 'P', '081555667044',
'farah.quinn@outlook.com', '2023-11-14'),
('3586454545940045', 'Lukman Sardi', 'L', '081966778045',
'lukman.sardi@gmail.com', '2023-11-15'),
('3180464646950046', 'Gita Gutawa', 'P', '083877889046',
'gita.gutawa@yahoo.com', '2023-11-16'),
('3282474747960047', 'Maulana Malik', 'L', '085788990047',
'maulana.malik@gmail.com', '2023-11-17'),
('3383484848970048', 'Hana Tajima', 'P', '081299001048',
'hana.tajima@gmail.com', '2023-11-18'),
('3480494949980049', 'Naufal Azhar', 'L', '089600112049',
'naufal.azhar@outlook.com', '2023-11-19'),
('3587505050990050', 'Intan Nuraini', 'P', '081311223050',
'intan.nuraini@gmail.com', '2023-11-20'),
('3181515151900051', 'Oki Setiana', 'P', '082122334051',
'oki.setiana@yahoo.com', '2023-11-21'),
('3283525252910052', 'Pratama Arhan', 'L', '087833445052',
'pratama.arhan@gmail.com', '2023-11-22'),
('3384535353920053', 'Kartika Putri', 'P', '081544556053',
'kartika.putri@gmail.com', '2023-11-23'),
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('3481545454930054', 'Rizki Febian', 'L', '081955667054',
'rizki.febian@outlook.com', '2023-11-24'),
('3588555555940055', 'Larasati', 'P', '083866778055',
'larasati.94@gmail.com', '2023-11-25'),
('3182565656950056', 'Surya Saputra', 'L', '085777889056',
'surya.saputra@yahoo.com', '2023-11-26'),
('3284575757960057', 'Melati Daeva', 'P', '081288990057',
'melati.daeva@gmail.com', '2023-11-27'),
('3385585858970058', 'Teguh Karya', 'L', '089699001058',
'teguh.karya@gmail.com', '2023-11-28'),
('3482595959980059', 'Nadia Vega', 'P', '081300112059',
'nadia.vega@outlook.com', '2023-11-29'),
('3589606060990060', 'Wawan Hendrawan', 'L', '082111223060',
'wawan.h@gmail.com', '2023-11-30'),
('3183616161900061', 'Olivia Jensen', 'P', '087822334061',
'olivia.jensen@yahoo.com', '2023-12-01'),
('3285626262910062', 'Zainal Abidin', 'L', '081533445062',
'zainal.abidin@gmail.com', '2023-12-02'),
('3386636363920063', 'Pratiwi Sudarmono', 'P', '081944556063',
'pratiwi.s@gmail.com', '2023-12-03'),
('3483646464930064', 'Ahmad Dhani', 'L', '083855667064',
'ahmad.dhani@outlook.com', '2023-12-04'),
('3590656565940065', 'Qisthi Nabilah', 'P', '085766778065',
'qisthi.nabilah@gmail.com', '2023-12-05'),
('3184666666950066', 'Dedi Mizwar', 'L', '081277889066',
'dedi.mizwar@yahoo.com', '2023-12-06'),
('3286676767960067', 'Ratih Purwasih', 'P', '089688990067',
'ratih.purwasih@gmail.com', '2023-12-07'),
('3387686868970068', 'Firmansyah', 'L', '081399001068',
'firmansyah.97@gmail.com', '2023-12-08'),
('3484696969980069', 'Salsabila', 'P', '082100112069',
'salsabila.98@outlook.com', '2023-12-09'),
('3591707070990070', 'Hadi Tjahjanto', 'L', '087811223070',
'hadi.tjahjanto@gmail.com', '2023-12-10'),
('3185717171900071', 'Tiara Andini', 'P', '081522334071',
'tiara.andini@yahoo.com', '2023-12-11'),
('3287727272910072', 'Indra Sjafri', 'L', '081933445072',
'indra.sjafri@gmail.com', '2023-12-12'),
('3388737373920073', 'Utami Dewi', 'P', '083844556073',
'utami.dewi@gmail.com', '2023-12-13'),
('3485747474930074', 'Junaidi', 'L', '085755667074',
'junaidi.93@outlook.com', '2023-12-14'),
('3592757575940075', 'Vina Panduwinata', 'P', '081266778075',
'vina.p@gmail.com', '2023-12-15'),
('3186767676950076', 'Kurniawan Dwi', 'L', '089677889076',
'kurniawan.dwi@yahoo.com', '2023-12-16'),
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('3288777777960077', 'Winda Viska', 'P', '081388990077',
'winda.viska@gmail.com', '2023-12-17'),
('3389787878970078', 'Ramadhan Sananta', 'L', '082199001078',
'ramadhan.s@gmail.com', '2023-12-18'),
('3486797979980079', 'Yulia Rachman', 'P', '087800112079',
'yulia.rachman@outlook.com', '2023-12-19'),
('3593808080990080', 'Wibowo', 'L', '081511223080',
'wibowo.99@gmail.com', '2023-12-20'),
('3187818181900081', 'Zahra Lari', 'P', '081922334081',
'zahra.lari@yahoo.com', '2023-12-21'),
('3289828282910082', 'Yanto Basna', 'L', '083833445082',
'yanto.basna@gmail.com', '2023-12-22'),
('3390838383920083', 'Aisyah Aqilah', 'P', '085744556083',
'aisyah.aqilah@gmail.com', '2023-12-23'),
('3487848484930084', 'Taufik Hidayat', 'L', '081255667084',
'taufik.h@outlook.com', '2023-12-24'),
('3594858585940085', 'Dinda Hauw', 'P', '089666778085',
'dinda.hauw@gmail.com', '2023-12-25'),
('3188868686950086', 'Bambang Pamungkas', 'L', '081377889086',
'bambang.p@yahoo.com', '2023-12-26'),
('3290878787960087', 'Mega Mustika', 'P', '082188990087',
'mega.mustika@gmail.com', '2023-12-27'),
('3391888888970088', 'Andik Vermansyah', 'L', '087899001088',
'andik.v@gmail.com', '2023-12-28'),
('3488898989980089', 'Novita Angie', 'P', '081500112089',
'novita.angie@outlook.com', '2023-12-29'),
('3595909090990090', 'Evan Dimas', 'L', '081911223090',
'evan.dimas@gmail.com', '2023-12-30'),
('3189919191900091', 'Maudy Ayunda', 'P', '083822334091',
'maudy.ayunda@yahoo.com', '2023-12-31'),
('3291929292910092', 'Reza Rahadian', 'L', '085733445092',
'reza.rahadian@gmail.com', '2024-01-01'),
('3392939393920093', 'Chelsea Islan', 'P', '081244556093',
'chelsea.islan@gmail.com', '2024-01-02'),
('3489949494930094', 'Vino Bastian', 'L', '089655667094',
'vino.bastian@outlook.com', '2024-01-03'),
('3596959595940095', 'Tara Basro', 'P', '081366778095',
'tara.basro@gmail.com', '2024-01-04'),
('3190969696950096', 'Chicco Jerikho', 'L', '082177889096',
'chicco.jerikho@yahoo.com', '2024-01-05'),
('3292979797960097', 'Pevita Pearce', 'P', '087888990097',
'pevita.pearce@gmail.com', '2024-01-06'),
('3393989898970098', 'Iko Uwais', 'L', '081599001098',
'iko.uwais@gmail.com', '2024-01-07'),
('3490999999980099', 'Adinia Wirasti', 'P', '081900112099',
'adinia.wirasti@outlook.com', '2024-01-08'),
```

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('3597000000990100', 'Joe Taslim', 'L', '083811223100',
'joe.taslim@gmail.com', '2024-01-09');


INSERT INTO kereta (id_kereta, nama_kereta, kapasitas) VALUES
('KA001', 'Argo Bromo Anggrek', 400),
('KA002', 'Argo Wilis', 400),
('KA003', 'Argo Lawu', 400),
('KA004', 'Argo Dwipangga', 400),
('KA005', 'Argo Sindoro', 400),
('KA006', 'Argo Muria', 400),
('KA007', 'Argo Parahyangan', 550),
('KA008', 'Argo Cheribon', 450),
('KA009', 'Argo Semeru', 450),
('KA010', 'Argo Merbabu', 450),
('KA011', 'Turangga', 450),
('KA012', 'Gajayana', 450),
('KA013', 'Bima', 400),
('KA014', 'Sembrani', 400),
('KA015', 'Taksaka', 450),
('KA016', 'Purwojaya', 450),
('KA017', 'Lodaya', 550),
('KA018', 'Mutiara Selatan', 600),
('KA019', 'Malabar', 650),
('KA020', 'Harina', 600),
('KA021', 'Gumarang', 650),
('KA022', 'Sancaka', 550),
('KA023', 'Ranggajati', 600),
('KA024', 'Wijayakusuma', 600),
('KA025', 'Mutiara Timur', 550),
('KA026', 'Majapahit', 700),
('KA027', 'Jayabaya', 650),
('KA028', 'Brantas', 750),
('KA029', 'Singasari', 700),
('KA030', 'Gaya Baru Malam', 800),
('KA031', 'Pasundan', 850),
('KA032', 'Kahuripan', 850),
('KA033', 'Logawa', 800),
('KA034', 'Kutojaya Utara', 800),
('KA035', 'Kutojaya Selatan', 850),
('KA036', 'Progo', 850),
('KA037', 'Senja Utama Solo', 600),
('KA038', 'Fajar Utama Solo', 600),
('KA039', 'Senja Utama Yogya', 650),
('KA040', 'Fajar Utama Yogya', 650),
('KA041', 'Bogowonto', 700),
('KA042', 'Gajahwong', 700),
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('KA043', 'Jaka Tingkir', 750),
('KA044', 'Bengawan', 850),
('KA045', 'Matarmaja', 850),
('KA046', 'Kertajaya', 1000),
('KA047', 'Airlangga', 850),
('KA048', 'Tawang Jaya', 850),
('KA049', 'Tawang Alun', 600),
('KA050', 'Sri Tanjung', 850),
('KA051', 'Probowangi', 600),
('KA052', 'Pandanwangi', 600),
('KA053', 'Kamandaka', 500),
('KA054', 'Joglosemarkerto', 500),
('KA055', 'Kaligung', 600),
('KA056', 'Kedung Sepur', 300),
('KA057', 'Blora Jaya', 500),
('KA058', 'Sribilah', 550),
('KA059', 'Putri Deli', 650),
('KA060', 'Sindang Marga', 500),
('KA061', 'Serelo', 650),
('KA062', 'Rajabasa', 650),
('KA063', 'Kuala Stabas', 550),
('KA064', 'Sibinuang', 600),
('KA065', 'Prambanan Ekspres', 500),
('KA066', 'Bathara Kresna', 150),
('KA067', 'Kalimaya', 700),
('KA068', 'Rangkas Jaya', 700),
('KA069', 'Langsam', 800),
('KA070', 'Banten Ekspres', 800),
('KA071', 'Walahar Ekspres', 850),
('KA072', 'Jatiluhur', 850),
('KA073', 'Cilamaya Ekspres', 850),
('KA074', 'Tegal Ekspres', 800),
('KA075', 'Maharani', 700),
('KA076', 'Pangrango', 400),
('KA077', 'Siliwangi', 500),
('KA078', 'Baturraden Ekspres', 550),
('KA079', 'Nusa Tembini', 550),
('KA080', 'Pandalungan', 600),
('KA081', 'Blambangan Ekspres', 600),
('KA082', 'Dharmawangsa', 700),
('KA083', 'Kertanegara', 600),
('KA084', 'Bangunkarta', 650),
('KA085', 'Mataram', 650),
('KA086', 'Manahan', 450),
('KA087', 'Sawunggalih', 650),
('KA088', 'Menoreh', 700),
('KA089', 'Ciremai', 650),
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('KA090', 'Ambarawa Ekspres', 600),
('KA091', 'Jayakarta', 850),
('KA092', 'Penataran', 800),
('KA093', 'Dhoho', 800),
('KA094', 'Tumapel', 800),
('KA095', 'Sancaka Utara', 550),
('KA096', 'Argo Jati', 450),
('KA097', 'Tawangmangu', 500),
('KA098', 'Macan Kumbang', 500),
('KA099', 'Sriwijaya', 550),
('KA100', 'Kintamani', 600);


INSERT INTO stasiun (kode_stasiun, nama_stasiun, kota,
wilayah_daop) VALUES
('GMR', 'Gambir', 'Jakarta', 'Daop 1 Jakarta'),
('PSE', 'Pasar Senen', 'Jakarta', 'Daop 1 Jakarta'),
('JAKK', 'Jakarta Kota', 'Jakarta', 'Daop 1 Jakarta'),
('JNG', 'Jatinegara', 'Jakarta', 'Daop 1 Jakarta'),
('MRI', 'Manggarai', 'Jakarta', 'Daop 1 Jakarta'),
('THB', 'Tanah Abang', 'Jakarta', 'Daop 1 Jakarta'),
('BKS', 'Bekasi', 'Bekasi', 'Daop 1 Jakarta'),
('CKR', 'Cikarang', 'Bekasi', 'Daop 1 Jakarta'),
('BOO', 'Bogor', 'Bogor', 'Daop 1 Jakarta'),
('DP', 'Depok', 'Depok', 'Daop 1 Jakarta'),
('RK', 'Rangkasbitung', 'Lebak', 'Daop 1 Jakarta'),
('SG', 'Serang', 'Serang', 'Daop 1 Jakarta'),
('MER', 'Merak', 'Cilegon', 'Daop 1 Jakarta'),
('KW', 'Karawang', 'Karawang', 'Daop 1 Jakarta'),
('CKP', 'Cikampek', 'Karawang', 'Daop 1 Jakarta'),
('BD', 'Bandung', 'Bandung', 'Daop 2 Bandung'),
('KAC', 'Kiaracondong', 'Bandung', 'Daop 2 Bandung'),
('CMI', 'Cimahi', 'Cimahi', 'Daop 2 Bandung'),
('PDL', 'Padalarang', 'Bandung Barat', 'Daop 2 Bandung'),
('CCL', 'Cicalengka', 'Bandung', 'Daop 2 Bandung'),
('PWK', 'Purwakarta', 'Purwakarta', 'Daop 2 Bandung'),
('GRT', 'Garut', 'Garut', 'Daop 2 Bandung'),
('CB', 'Cibatu', 'Garut', 'Daop 2 Bandung'),
('CPD', 'Cipeundeuy', 'Garut', 'Daop 2 Bandung'),
('TSM', 'Tasikmalaya', 'Tasikmalaya', 'Daop 2 Bandung'),
('BJR', 'Banjar', 'Banjar', 'Daop 2 Bandung'),
('CN', 'Cirebon', 'Cirebon', 'Daop 3 Cirebon'),
('CNP', 'Cirebon Prujakan', 'Cirebon', 'Daop 3 Cirebon'),
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('SMC', 'Semarang Poncol', 'Semarang', 'Daop 4 Semarang'),
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('NBO', 'Ngrombo', 'Grobogan', 'Daop 4 Semarang'),
('CU', 'Cepu', 'Blora', 'Daop 4 Semarang'),
('PWT', 'Purwokerto', 'Banyumas', 'Daop 5 Purwokerto'),
('KYA', 'Kroya', 'Cilacap', 'Daop 5 Purwokerto'),
('CP', 'Cilacap', 'Cilacap', 'Daop 5 Purwokerto'),
('MA', 'Maos', 'Cilacap', 'Daop 5 Purwokerto'),
('BMA', 'Bumiayu', 'Brebes', 'Daop 5 Purwokerto'),
('SLW', 'Slawi', 'Tegal', 'Daop 5 Purwokerto'),
('GB', 'Gombong', 'Kebumen', 'Daop 5 Purwokerto'),
('KA', 'Karanganyar', 'Kebumen', 'Daop 5 Purwokerto'),
('KM', 'Kebumen', 'Kebumen', 'Daop 5 Purwokerto'),
('KTA', 'Kutoarjo', 'Purworejo', 'Daop 5 Purwokerto'),
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('WT', 'Wates', 'Kulon Progo', 'Daop 6 Yogyakarta'),
('BBN', 'Brambanan', 'Klaten', 'Daop 6 Yogyakarta'),
('KT', 'Klaten', 'Klaten', 'Daop 6 Yogyakarta'),
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('SLO', 'Solo Balapan', 'Surakarta', 'Daop 6 Yogyakarta'),
('SK', 'Solo Jebres', 'Surakarta', 'Daop 6 Yogyakarta'),
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('LW', 'Lawang', 'Malang', 'Daop 8 Surabaya'),
('MR', 'Mojokerto', 'Mojokerto', 'Daop 8 Surabaya'),
('LMG', 'Lamongan', 'Lamongan', 'Daop 8 Surabaya'),
('BBT', 'Babat', 'Lamongan', 'Daop 8 Surabaya'),
('BJ', 'Bojonegoro', 'Bojonegoro', 'Daop 8 Surabaya'),
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('RBP', 'Rambipuji', 'Jember', 'Daop 9 Jember'),
('TGL', 'Tanggul', 'Jember', 'Daop 9 Jember'),
('KLT', 'Kalisat', 'Jember', 'Daop 9 Jember'),
('KK', 'Klakah', 'Lumajang', 'Daop 9 Jember'),
('PB', 'Probolinggo', 'Probolinggo', 'Daop 9 Jember'),
('PS', 'Pasuruan', 'Pasuruan', 'Daop 9 Jember'),
('KBR', 'Kalibaru', 'Banyuwangi', 'Daop 9 Jember'),
('RGP', 'Rogojampi', 'Banyuwangi', 'Daop 9 Jember'),
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INSERT INTO tiket (id_tiket, kode_booking, id_penumpang,
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(5, 'P5R8T1', 5, 88, 5, '2A', '2024-02-05 07:12:07',
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(43, 'O8I6U4', 43, 49, 2, '10B', '2024-02-06 19:59:27', 'Lunas',
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(44, 'Y2T1R9', 44, 16, 6, '4C', '2024-02-07 08:28:13', 'Lunas',
'OVO'),
(45, 'E5W7Q9', 45, 61, 1, '11D', '2024-02-08 17:57:37',
'Dibatalkan', 'Indomaret'),
(46, 'A3S5D7', 46, 38, 5, '7A', '2024-02-09 05:18:48', 'Lunas',
'ShopeePay'),
(47, 'Z6X4C2', 47, 94, 4, '15B', '2024-02-10 03:21:16', 'Lunas',
'BCA Virtual Account'),
```

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(48, 'V1B3N5', 48, 9, 3, '1C', '2024-02-11 08:56:36', 'Lunas',
'BRI Virtual Account'),
(49, 'M5M7M9', 49, 68, 2, '9D', '2024-02-12 05:18:53',
'Dibatalkan', 'QRIS'),
(50, 'K2J4H6', 50, 21, 6, '5A', '2024-02-13 13:02:34', 'Lunas',
'GoPay'),
(51, 'G7F9D1', 51, 79, 1, '12B', '2024-02-14 18:04:05', 'Lunas',
'Mandiri VA'),
(52, 'S4A6P8', 52, 42, 5, '8C', '2024-02-15 05:56:56', 'Lunas',
'BCA Virtual Account'),
(53, 'O3I5U7', 53, 13, 4, '2D', '2024-02-16 22:45:03', 'Lunas',
'QRIS'),
(54, 'Y9T8R1', 54, 57, 3, '14A', '2024-02-17 23:42:36', 'Lunas',
'OVO'),
(55, 'E2W4Q6', 55, 34, 2, '10B', '2024-02-18 22:52:33',
'Dibatalkan', 'Indomaret'),
(56, 'A8S9D1', 56, 89, 6, '4C', '2024-02-19 03:59:32', 'Lunas',
'ShopeePay'),
(57, 'Z5X7C9', 57, 4, 1, '11D', '2024-02-20 18:11:38', 'Lunas',
'BCA Virtual Account'),
(58, 'V2B4N6', 58, 63, 5, '7A', '2024-02-21 15:20:42',
'Dibatalkan', 'BRI Virtual Account'),
(59, 'M8M1M3', 59, 26, 4, '15B', '2024-02-22 13:37:30', 'Lunas',
'QRIS'),
(60, 'K4J5H7', 60, 83, 3, '1C', '2024-02-23 15:37:14', 'Lunas',
'GoPay'),
(61, 'G9F1D2', 61, 46, 2, '9D', '2024-02-24 07:49:11', 'Lunas',
'Mandiri VA'),
(62, 'S5A7P9', 62, 17, 6, '5A', '2024-02-25 12:57:17', 'Lunas',
'BCA Virtual Account'),
(63, 'O2I4U6', 63, 71, 1, '12B', '2024-02-26 14:03:33', 'Lunas',
'QRIS'),
(64, 'Y8T9R2', 64, 30, 5, '8C', '2024-02-27 13:13:21', 'Lunas',
'OVO'),
(65, 'E6W1Q3', 65, 87, 4, '2D', '2024-02-28 03:39:52',
'Dibatalkan', 'Indomaret'),
(66, 'A4S6D8', 66, 1, 3, '14A', '2024-03-01 08:12:55', 'Lunas',
'ShopeePay'),
(67, 'Z7X9C1', 67, 59, 2, '10B', '2024-03-02 23:10:53', 'Lunas',
'BCA Virtual Account'),
(68, 'V3B5N7', 68, 20, 6, '4C', '2024-03-03 22:00:13',
'Dibatalkan', 'BRI Virtual Account'),
(69, 'M1M9M8', 69, 75, 1, '11D', '2024-03-04 02:56:18', 'Lunas',
'QRIS'),
(70, 'K6J7H9', 70, 37, 5, '7A', '2024-03-05 16:35:53', 'Lunas',
'GoPay'),
```

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(71, 'G2F8D3', 71, 93, 4, '15B', '2024-03-06 15:31:42', 'Lunas',
'Mandiri VA'),
(72, 'S8A1P4', 72, 10, 3, '1C', '2024-03-07 23:47:59', 'Lunas',
'BCA Virtual Account'),
(73, 'O5I7U9', 73, 64, 2, '9D', '2024-03-08 14:04:33', 'Lunas',
'QRIS'),
(74, 'Y1T3R5', 74, 28, 6, '5A', '2024-03-09 13:02:34', 'Lunas',
'OVO'),
(75, 'E7W9Q2', 75, 82, 1, '12B', '2024-02-05 18:10:31',
'Dibatalkan', 'Indomaret'),
(76, 'A5S7D9', 76, 40, 5, '8C', '2024-02-10 01:45:38', 'Lunas',
'ShopeePay'),
(77, 'Z3X1C4', 77, 98, 4, '2D', '2024-02-15 21:33:07', 'Lunas',
'BCA Virtual Account'),
(78, 'V8B2N5', 78, 6, 3, '14A', '2024-02-20 21:11:14',
'Dibatalkan', 'BRI Virtual Account'),
(79, 'M4M6M8', 79, 60, 2, '10B', '2024-02-25 15:25:12', 'Lunas',
'QRIS'),
(80, 'K9J1H2', 80, 24, 6, '4C', '2024-03-01 17:08:28', 'Lunas',
'GoPay'),
(81, 'G5F7D9', 81, 78, 1, '11D', '2024-03-02 23:23:04', 'Lunas',
'Mandiri VA'),
(82, 'S2A4P6', 82, 36, 5, '7A', '2024-03-03 04:46:14', 'Lunas',
'BCA Virtual Account'),
(83, 'O9I2U4', 83, 90, 4, '15B', '2024-03-04 22:53:40', 'Lunas',
'QRIS'),
(84, 'Y6T8R1', 84, 15, 3, '1C', '2024-03-05 16:22:51',
'Dibatalkan', 'OVO'),
(85, 'E3W5Q7', 85, 69, 2, '9D', '2024-03-06 04:21:12', 'Lunas',
'Indomaret'),
(86, 'A1S9D2', 86, 32, 6, '5A', '2024-03-07 22:09:56', 'Lunas',
'ShopeePay'),
(87, 'Z8X2C5', 87, 86, 1, '12B', '2024-03-08 06:26:27', 'Lunas',
'BCA Virtual Account'),
(88, 'V5B7N9', 88, 43, 5, '8C', '2024-03-09 23:30:35',
'Dibatalkan', 'BRI Virtual Account'),
(89, 'M7M9M1', 89, 100, 4, '2D', '2024-02-12 10:25:46', 'Lunas',
'QRIS'),
(90, 'K3J5H8', 90, 11, 3, '14A', '2024-02-14 15:55:41', 'Lunas',
'GoPay'),
(91, 'G8F2D4', 91, 65, 2, '10B', '2024-02-16 15:07:43', 'Lunas',
'Mandiri VA'),
(92, 'S1A6P9', 92, 25, 6, '4C', '2024-02-18 06:05:41', 'Lunas',
'BCA Virtual Account'),
(93, 'O4I8U2', 93, 80, 1, '11D', '2024-02-20 04:56:05', 'Lunas',
'QRIS'),
```

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(94, 'Y7T1R4', 94, 38, 5, '7A', '2024-02-22 15:54:38',
'Dibatalkan', 'OVO'),
(95, 'E1W6Q8', 95, 96, 4, '15B', '2024-02-24 11:24:39', 'Lunas',
'Indomaret'),
(96, 'A6S2D5', 96, 17, 3, '1C', '2024-02-26 01:21:35', 'Lunas',
'ShopeePay'),
(97, 'Z2X8C1', 97, 70, 2, '9D', '2024-02-28 21:43:56', 'Lunas',
'BCA Virtual Account'),
(98, 'V9B4N6', 98, 30, 6, '5A', '2024-03-01 12:43:05',
'Dibatalkan', 'BRI Virtual Account'),
(99, 'M6M1M5', 99, 85, 1, '12B', '2024-03-03 06:02:26', 'Lunas',
'QRIS'),
(100, 'K5J8H2', 100, 47, 5, '8C', '2024-03-05 01:57:38', 'Lunas',
'GoPay'),
(101, 'BKA99Z', 1, 5, 3, '12A', '2026-02-26 12:13:24', 'Pending',
'QRIS');


CREATE TABLE kursi (
    id_kursi     INT PRIMARY KEY AUTO_INCREMENT,
    id_jadwal    INT NOT NULL,
    nomor_gerbong INT NOT NULL,
    nomor_kursi  VARCHAR(5) NOT NULL,
    status_kursi ENUM('Tersedia', 'Terisi') DEFAULT 'Tersedia',

    FOREIGN KEY (id_jadwal) REFERENCES jadwal(id_jadwal)
);

INSERT INTO kursi (id_jadwal, nomor_gerbong, nomor_kursi,
status_kursi) VALUES
-- Data yang sinkron dengan tabel Tiket (Status: Terisi)
(1, 2, '12A', 'Terisi'),
(2, 1, '4B', 'Terisi'),
(3, 4, '8C', 'Terisi'),
(4, 3, '15D', 'Terisi'),
(6, 2, '10B', 'Terisi'),
(7, 6, '14C', 'Terisi'),
(9, 3, '7D', 'Terisi'),
(10, 4, '11A', 'Terisi'),
(11, 2, '3B', 'Terisi'),
(13, 1, '16D', 'Terisi'),
(14, 3, '5A', 'Terisi'),
(15, 2, '12B', 'Terisi'),
(16, 4, '8C', 'Terisi'),
(17, 6, '2D', 'Terisi'),
(19, 5, '10B', 'Terisi'),
(20, 3, '4C', 'Terisi'),
```

```
(22, 4, '7A', 'Terisi'),
(23, 1, '15B', 'Terisi'),
(24, 6, '1C', 'Terisi'),
(26, 5, '5A', 'Terisi'),
(27, 2, '12B', 'Terisi'),
(28, 4, '8C', 'Terisi'),
(29, 1, '2D', 'Terisi'),
(30, 3, '14A', 'Terisi'),
(32, 2, '4C', 'Terisi'),
(33, 5, '11D', 'Terisi'),
(35, 4, '15B', 'Terisi'),
(36, 3, '1C', 'Terisi'),
(37, 2, '9D', 'Terisi'),
(38, 6, '5A', 'Terisi'),
(39, 1, '12B', 'Terisi'),
(41, 4, '2D', 'Terisi'),
(42, 3, '14A', 'Terisi'),
(43, 2, '10B', 'Terisi'),
(44, 6, '4C', 'Terisi'),
(46, 5, '7A', 'Terisi'),
(47, 4, '15B', 'Terisi'),
(48, 3, '1C', 'Terisi'),
(50, 6, '5A', 'Terisi'),

-- Data Kursi Tambahan (Status: Tersedia)
(1, 1, '1A', 'Tersedia'),
(1, 1, '1B', 'Tersedia'),
(1, 1, '1C', 'Tersedia'),
(1, 1, '1D', 'Tersedia'),
(2, 1, '2A', 'Tersedia'),
(2, 1, '2B', 'Tersedia'),
(2, 1, '2C', 'Tersedia'),
(3, 2, '3A', 'Tersedia'),
(3, 2, '3B', 'Tersedia'),
(4, 1, '5A', 'Tersedia'),
(4, 1, '5B', 'Tersedia'),
(5, 1, '1A', 'Tersedia'),
(5, 1, '1B', 'Tersedia'),
(6, 1, '10A', 'Tersedia'),
(7, 2, '14A', 'Tersedia'),
(8, 1, '1A', 'Tersedia'),
(9, 1, '7A', 'Tersedia'),
(10, 1, '11B', 'Tersedia'),
(11, 1, '3A', 'Tersedia'),
(12, 1, '9A', 'Tersedia'),
(13, 2, '16A', 'Tersedia'),
(14, 2, '5B', 'Tersedia'),
```

```
(15, 1, '12A', 'Tersedia'),
(16, 1, '8A', 'Tersedia'),
(17, 1, '2A', 'Tersedia'),
(18, 1, '14B', 'Tersedia'),
(19, 1, '10A', 'Tersedia'),
(20, 1, '4A', 'Tersedia'),
(21, 1, '11A', 'Tersedia'),
(22, 1, '7B', 'Tersedia'),
(23, 2, '15A', 'Tersedia'),
(24, 2, '1A', 'Tersedia'),
(25, 1, '9A', 'Tersedia'),
(51, 1, '1A', 'Tersedia'),
(51, 1, '1B', 'Tersedia'),
(52, 2, '8A', 'Tersedia'),
(53, 3, '2A', 'Tersedia'),
(54, 1, '14B', 'Tersedia'),
(55, 1, '10A', 'Tersedia'),
(56, 1, '4A', 'Tersedia'),
(57, 1, '11A', 'Tersedia'),
(58, 2, '7B', 'Tersedia'),
(59, 3, '15A', 'Tersedia'),
(60, 1, '1A', 'Tersedia'),
(61, 1, '9B', 'Tersedia'),
(62, 2, '5B', 'Tersedia'),
(63, 1, '12A', 'Tersedia'),
(64, 1, '8A', 'Tersedia'),
(65, 3, '2A', 'Tersedia'),
(66, 1, '14B', 'Tersedia'),
(67, 1, '10A', 'Tersedia'),
(68, 2, '4A', 'Tersedia'),
(69, 1, '11A', 'Tersedia'),
(70, 1, '7B', 'Tersedia'),
(71, 2, '15A', 'Tersedia'),
(72, 1, '1A', 'Tersedia'),
(73, 3, '9A', 'Tersedia'),
(74, 1, '5B', 'Tersedia'),
(75, 1, '12A', 'Tersedia');
```

STORED PROCEDURE

1. **Stored Procedure: Hapus Penumpang**

   Prosedur ini akan menghapus data penumpang berdasarkan id_penumpang. Sebelum menghapus data penumpang, prosedur ini akan menghapus semua riwayat tiket yang terkait dengan penumpang tersebut agar tidak terjadi *Foreign Key Error*.

```
DELIMITER //

CREATE PROCEDURE sp_delete_penumpang(
    IN p_id_penumpang INT
)
BEGIN
          DELETE   FROM   tiket   WHERE   id_penumpang   =
p_id_penumpang;
         DELETE  FROM  penumpang  WHERE  id_penumpang  =
p_id_penumpang;
END //

DELIMITER ;
Cara Penggunaan :
CALL sp_delete_penumpang(5):
```

2. **Stored Procedure: Hapus Jadwal Kereta**

   Prosedur ini digunakan untuk menghapus jadwal perjalanan. Sama seperti sebelumnya, karena id_jadwal digunakan di tabel tiket, kita perlu membersihkan data tiket yang sudah dipesan pada jadwal tersebut sebelum jadwalnya bisa dihapus.

```
DELIMITER //

CREATE PROCEDURE sp_delete_jadwal(
    IN p_id_jadwal INT
)
BEGIN
    DELETE FROM tiket WHERE id_jadwal = p_id_jadwal;
    DELETE FROM jadwal WHERE id_jadwal = p_id_jadwal;
END //

DELIMITER ;
Cara Penggunaan :
CALL sp_delete_jadwal(10):
```

3. **Stored Procedure: Menampilkan Detail Tiket**

Stored procedure ini bertujuan untuk mempermudah pencarian data tiket secara cepat dan terpusat hanya dengan menggunakan kode booking tanpa perlu menjalankan query kompleks secara manual.

```
DELIMITER //

CREATE PROCEDURE sp_TiketPerKode(IN KodeBooking
VARCHAR(20))
BEGIN
    SELECT
        tiket.kode_booking,
        penumpang.nama_lengkap AS nama_penumpang,
        penumpang.no_telepon,
        kereta.nama_kereta,
        stasiun_asal.nama_stasiun AS stasiun_asal,
        stasiun_tujuan.nama_stasiun AS stasiun_tujuan,
        jadwal.waktu_berangkat,
        jadwal.waktu_tiba,
        jadwal.harga,
        tiket.nomor_gerbong,
        tiket.nomor_kursi,
        tiket.status_pembayaran,
        tiket.metode_pembayaran
    FROM tiket
    JOIN penumpang
        ON tiket.id_penumpang = penumpang.id_penumpang
    JOIN jadwal
        ON tiket.id_jadwal = jadwal.id_jadwal
    JOIN kereta
        ON jadwal.id_kereta = kereta.id_kereta
    JOIN stasiun AS stasiun_asal
        ON jadwal.id_stasiun_asal =
stasiun_asal.id_stasiun
    JOIN stasiun AS stasiun_tujuan
        ON jadwal.id_stasiun_tujuan =
stasiun_tujuan.id_stasiun
    WHERE tiket.kode_booking = KodeBooking;
END //

DELIMITER ;

CALL sp_TiketPerKode(X7A9KQ));
```

output :

```
MariaDB [kereta]> CALL sp_TiketPerKode('X7A9KQ');
+--------------+----------------+--------------+----------------+--------------+-------------------+-----------------
---------+---------------------+--------+---------------+-------------+-------------------+-----------------
----+
| kode_booking | nama_penumpang | no_telepon   | nama_kereta    | stasiun_asal | stasiun_tujuan | waktu_beran
gkat     | waktu_tiba          | harga  | nomor_gerbong | nomor_kursi | status_pembayaran | metode_pembayara
n   |
+--------------+----------------+--------------+----------------+--------------+-------------------+-----------------
---------+---------------------+--------+---------------+-------------+-------------------+-----------------
----+
| X7A9KQ       | Budi Santoso   | 081234567001 | Argo Cheribon | Gambir       | Cirebon         | 2024-03-10
08:20:00 | 2024-03-10 11:20:00 | 200000 |             2 | 12A         | Lunas             | BCA Virtual Acco
unt |
+--------------+----------------+--------------+----------------+--------------+-------------------+-----------------
---------+---------------------+--------+---------------+-------------+-------------------+-----------------
----+
1 row in set (0.001 sec)

Query OK, 0 rows affected (0.011 sec)
```

4. **Stored Procedure : Menampilkan penumpang yang di hari tersebut**

Stored procedure ini bertujuan untuk membantu administrator dalam melihat daftar penumpang perjalanan kereta pada hari tertentu secara cepat tanpa perlu menuliskan query kompleks secara manual.

```
DELIMITER //

CREATE PROCEDURE sp_PenumpangPerHari(IN TanggalDipilih
DATE)
BEGIN
    SELECT
        tiket.kode_booking,
        penumpang.nama_lengkap,
        penumpang.no_telepon,
        kereta.nama_kereta,
        stasiun_asal.nama_stasiun AS stasiun_asal,
        stasiun_tujuan.nama_stasiun AS stasiun_tujuan,
        jadwal.waktu_berangkat,
        tiket.nomor_gerbong,
        tiket.nomor_kursi
    FROM tiket
    JOIN penumpang
        ON tiket.id_penumpang = penumpang.id_penumpang
    JOIN jadwal
        ON tiket.id_jadwal = jadwal.id_jadwal
    JOIN kereta
        ON jadwal.id_kereta = kereta.id_kereta
    JOIN stasiun AS stasiun_asal
        ON jadwal.id_stasiun_asal =
stasiun_asal.id_stasiun
    JOIN stasiun AS stasiun_tujuan
```

```
        ON jadwal.id_stasiun_tujuan =
stasiun_tujuan.id_stasiun
    WHERE DATE(jadwal.waktu_berangkat) = TanggalDipilih;
END //

DELIMITER ;

 CALL sp_PenumpangPerHari('2024-03-10');
```

Output:

```
MariaDB [kereta]> CALL sp_PenumpangPerHari('2024-03-10');
```

kode_booking	nama_lengkap	no_telepon	nama_kereta	stasiun_asal	stasiun_tujuan	waktu_berangkat	nomor_gerbong	nomor_kursi
X7A9KQ	Budi Santoso	081234567001	Argo Cheribon	Gambir	Cirebon	2024-03-10 08:20:00	2	12A
M4V2P9	Siti Aminah	085678900002	Lodaya	Bandung	Solo Balapan	2024-03-10 06:50:00	1	4B
Q2W4E6	Sri Rahayu	082233445006	Ranggajati	Cirebon	Jember	2024-03-10 05:30:00	2	10B
M3M5M7	Eko Saputra	081900112009	Kertajaya	Pasar Senen	Surabaya Pasarturi	2024-03-10 14:10:00	3	7D
K8J6H4	Rini Susanti	083811223010	Argo Muria	Gambir	Semarang Tawang	2024-03-10 07:00:00	4	11A
G2F4D6	Dwi Yulianto	085722334011	Bima	Gambir	Surabaya Gubeng	2024-03-10 17:00:00	2	3B
S1A3P5	Nur Hidayah	081233445012	Kutojaya Utara	Pasar Senen	Purwokerto	2024-03-10 05:05:00	5	9C
E1W3Q5	Anton Syahputra	082166778015	Harina	Bandung	Surabaya Pasarturi	2024-03-10 20:00:00	2	12B
A2S4D6	Ratna Kumala	087877889016	Sri Tanjung	Lempuyangan	Ketapang	2024-03-10 07:00:00	4	8C
V6B5N4	Fitriani	081999001018	Senja Utama Yogya	Pasar Senen	Yogyakarta	2024-03-10 19:10:00	1	14A
M1M2M3	Iwan Fals	083800112019	Gumarang	Pasar Senen	Surabaya Pasarturi	2024-03-10 15:50:00	5	10B
G6F5D4	Arif Rahman	081222334021	Argo Lawu	Gambir	Solo Balapan	2024-03-10 20:00:00	2	11D
S3A2P1	Rina Marlina	089633445022	Jaka Tingkir	Pasar Senen	Solo Jebres	2024-03-10 12:00:00	4	7A
O7I8U9	Joko Susilo	081344556023	Taksaka	Gambir	Yogyakarta	2024-03-10 09:00:00	1	15B
Y1T2R3	Eka Fitri	082155667024	Senja Utama Solo	Pasar Senen	Solo Balapan	2024-03-10 19:40:00	6	1C
Z2X1C3	Rahmat Hidayat	081988990027	Tawang Jaya	Pasar Senen	Semarang Poncol	2024-03-10 23:15:00	2	12B
V4B6N8	Nita Amelia	083899001028	Purwojaya	Gambir	Purwokerto	2024-03-10 14:30:00	4	8C
K1J2H3	Wulan Guritno	081211223030	Argo Dwipangga	Gambir	Solo Balapan	2024-03-10 08:45:00	3	14A
S9A8P7	Yuni Shara	081333445032	Gajayana	Gambir	Malang	2024-03-10 18:30:00	2	4C
O1I2U3	Dimas Anggara	082144556033	Bogowonto	Pasar Senen	Lempuyangan	2024-03-10 21:45:00	5	11D
E9W8Q7	Fadli Zon	081566778035	Argo Merbabu	Semarang Tawang	Gambir	2024-03-10 14:00:00	4	15B
Z4X6C8	Gilang Dirga	083888990037	Mutiara Selatan	Bandung	Surabaya Gubeng	2024-03-10 20:30:00	2	9D
G1F3D5	Iqbal Ramadhan	081322334041	Sembrani	Gambir	Surabaya Pasarturi	2024-03-10 19:30:00	4	2D
O8I6U4	Kevin Julio	087844556043	Mutiara Timur	Surabaya Gubeng	Ketapang	2024-03-10 08:30:00	2	10B
E5W7Q9	Lukman Sardi	081966778045	Pasundan	Kiaracondong	Surabaya Gubeng	2024-03-10 10:15:00	1	11D
V1B3N5	Hana Tajima	081299001048	Argo Sindoro	Semarang Tawang	Gambir	2024-03-10 06:00:00	3	1C
K2J4H6	Intan Nuraini	081311223050	Turangga	Bandung	Surabaya Gubeng	2024-03-10 18:00:00	6	5A
G7F9D1	Oki Setiana	082122334051	Fajar Utama Yogya	Pasar Senen	Yogyakarta	2024-03-10 06:40:00	1	12B
O3I5U7	Kartika Putri	081544556053	Argo Parahyangan	Gambir	Bandung	2024-03-10 06:30:00	4	2D
Y9T8R1	Rizki Febian	081955667054	Singasari	Pasar Senen	Malang	2024-03-10 20:30:00	3	14A
A8S9D1	Surya Saputra	085777889056	Matarmaja	Pasar Senen	Malang	2024-03-10 10:20:00	6	4C
V2B4N6	Teguh Karya	089699001058	Kahuripan	Kiaracondong	Malang	2024-03-10 23:10:00	5	7A
K4J5H7	Wawan Hendrawan	082111223060	Gajahwong	Pasar Senen	Lempuyangan	2024-03-10 07:30:00	3	1C
S5A7P9	Zainal Abidin	081533445062	Argo Semeru	Gambir	Surabaya Gubeng	2024-03-10 06:15:00	6	5A
O2I4U6	Pratiwi Sudarmono	081944556063	Progo	Pasar Senen	Lempuyangan	2024-03-10 22:30:00	1	12B
E6W1Q3	Qisthi Nabilah	085766778065	Bengawan	Pasar Senen	Solo Jebres	2024-03-10 06:00:00	4	2D
A4S6D8	Dedi Mizwar	081277889066	Argo Bromo Anggrek	Gambir	Surabaya Pasarturi	2024-03-10 08:00:00	3	14A
Z7X9C1	Ratih Purwasih	089688990067	Gaya Baru Malam	Pasar Senen	Surabaya Gubeng	2024-03-10 11:00:00	2	10B
M1M9M8	Salsabila	082100112069	Fajar Utama Solo	Pasar Senen	Solo Balapan	2024-03-10 05:55:00	1	11D
K6J7H9	Hadi Tjahjanto	087811223070	Malabar	Bandung	Malang	2024-03-10 17:15:00	5	7A
G2F8D3	Tiara Andini	081522334071	Airlangga	Pasar Senen	Surabaya Pasarturi	2024-03-10 11:15:00	4	15B
Y6T8R1	Taufik Hidayat	081255667084	Argo Cheribon	Gambir	Cirebon	2024-03-10 08:20:00	3	1C
E3W5Q7	Dinda Hauw	089666778085	Kutojaya Selatan	Kiaracondong	Purwokerto	2024-03-10 22:05:00	2	9D
V5B7N9	Andik Vermansyah	087899001088	Sancaka	Yogyakarta	Surabaya Gubeng	2024-03-10 06:45:00	5	8C
K3J5H8	Evan Dimas	081911223090	Argo Muria	Gambir	Semarang Tawang	2024-03-10 07:00:00	3	14A
G8F2D4	Maudy Ayunda	083822334091	Logawa	Purwokerto	Jember	2024-03-10 05:30:00	2	10B
S1A6P9	Reza Rahadian	085733445092	Bima	Gambir	Surabaya Gubeng	2024-03-10 17:00:00	6	4C
A6S2D5	Chicco Jerikho	082177889096	Argo Semeru	Gambir	Surabaya Gubeng	2024-03-10 06:15:00	3	1C
M6M1M5	Adinia Wirasti	081900112099	Jaka Tingkir	Pasar Senen	Solo Jebres	2024-03-10 12:00:00	1	12B
K5J8H2	Joe Taslim	083811223100	Wijayakusuma	Purwokerto	Ketapang	2024-03-10 15:20:00	5	8C

CONSTRAINTS

1. NOT NULL

Menspesifikasikan bahwa sebuah kolom tidak boleh berisi NULL. Contohnya sebagai berikut:

```
CREATE TABLE kereta (
    id_kereta VARCHAR(5) PRIMARY KEY,
    nama_kereta VARCHAR(50) NOT NULL,
    kapasitas INT NOT NULL
);
```

Penjelasan: nama_kereta dan juga kapasitas wajib diisi dan tidak boleh berisi NULL.

```
ALTER TABLE penumpang
MODIFY tanggal_daftar DATE NOT NULL;
```

Penjelasan: mengubah kolom tanggal_daftar pada tabel penumpang menjadi NOT NULL.

2. UNIQUE

Melarang nilai kembar atau sama dan memastikan data unik di seluruh kolom. Contohnya sebagai berikut:

```
CREATE TABLE stasiun (
    id_stasiun INT PRIMARY KEY AUTO_INCREMENT,
    kode_stasiun VARCHAR(10) UNIQUE NOT NULL,
    nama_stasiun VARCHAR(100) NOT NULL,
    kota VARCHAR(50) NOT NULL,
    wilayah_daop VARCHAR(20)
);
```

Penjelasan: pada kode_stasiun harus berisi data yang unik.

```
ALTER TABLE stasiun
ADD CONSTRAINT unique_nama_stasiun UNIQUE (nama_stasiun);
```

Penjelasan: menambahkan constraint UNIQUE pada kolom nama_stasiun di tabel stasiun agar tidak ada nama stasiun yang sama.

3. PRIMARY KEY

Identitas dari tabel yang hanya satu per tabel dan bisa digabung beberapa kolom. Contohnya sebagai berikut:

```
CREATE TABLE kereta (
    id_kereta VARCHAR(5) PRIMARY KEY,
    nama_kereta VARCHAR(50) NOT NULL,
    kapasitas INT NOT NULL
);
```

Penjelasan: pada tabel kereta, id_kereta menjadi Primary Key, sehingga harus unique dan tidak boleh NULL.

4. REFERENCES/FOREIGN KEY

Memastikan data anak(child) selalu memiliki induk(parent) yang valid. Mencegah data yatim piatu. Contohnya sebagai berikut:

```
CREATE TABLE jadwal (
id_jadwal INT AUTO_INCREMENT PRIMARY KEY,
id_kereta VARCHAR(5) NOT NULL,
    id_stasiun_asal INT NOT NULL,
    id_stasiun_tujuan INT NOT NULL,
    waktu_berangkat DATETIME NOT NULL,
    waktu_tiba DATETIME NOT NULL,
    harga INT NOT NULL,

FOREIGN KEY (id_kereta) REFERENCES kereta(id_kereta),
FOREIGN KEY (id_stasiun_asal) REFERENCES
stasiun(id_stasiun),
FOREIGN KEY (id_stasiun_tujuan) REFERENCES
stasiun(id_stasiun)
);
```

Penjelasan: Constraint REFERENCES digunakan untuk menghubungkan tabel jadwal dengan tabel lain, yaitu tabel kereta dan tabel stasiun.

5. CHECK

Menspesifikasikan kondisi logika. Jika nilai diluar range, input ditolak demi keamanan data.

```
ALTER TABLE jadwal
ADD CONSTRAINT check_harga
CHECK (harga >= 0);
```

Penjelasan: memastikan harga tiket tidak boleh kurang dari 0.

6. DEFAULT

Menspesifikasikan nilai standar ketika kolom tidak diisi user. Contohnya sebagai berikut:

```
ALTER TABLE tiket
MODIFY status_pembayaran VARCHAR(20) DEFAULT 'Pending';
```

Penjelasan: mengubah struktur tabel tiket kolom status_pembayaran yang jika tidak diisi otomatis : “Pending”.

TRIGGERS

1. INSERT TRIGGERS

Mencegah penambahan jadwal baru jika kereta yang sama sudah memiliki jadwal lain di rentang waktu yang bertabrakan.
Kondisi tabrakan waktu adalah dimana waktu berangkat jadwal baru lebih kecil dari waktu tiba jadwal lama dan waktu tiba jadwal baru lebih besar dari waktu berangkat jadwal lama.

```
CREATE TRIGGER cegah_tabrakan
BEFORE INSERT ON jadwal
FOR EACH ROW
BEGIN
    IF EXISTS (
        SELECT 1
        FROM jadwal
        WHERE id_kereta = NEW.id_kereta
          AND NEW.waktu_berangkat < waktu_tiba
          AND NEW.waktu_tiba > waktu_berangkat
    ) THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'ERROR: Jadwal bertabrakan!';
    END IF;
END$$
DELIMITER ;
```

2. UPDATE TRIGGERS

Saat status tiket berubah jadi lunas maka status kursi di tabel kursi otomatis jadi terisi. Dan sebaliknya, jika status tiket berubah jadi dibatalkan maka kursi kembali tersedia.

```
DELIMITER $$

CREATE TRIGGER update_kursi
AFTER UPDATE ON tiket
FOR EACH ROW
BEGIN
    IF NEW.status_pembayaran = 'Lunas'
       AND OLD.status_pembayaran <> 'Lunas' THEN

        UPDATE kursi
        SET status_kursi = 'Terisi'
        WHERE id_jadwal     = NEW.id_jadwal
          AND nomor_gerbong = NEW.nomor_gerbong
          AND nomor_kursi   = NEW.nomor_kursi;
    END IF;

    IF NEW.status_pembayaran = 'Dibatalkan'
       AND OLD.status_pembayaran <> 'Dibatalkan' THEN

        UPDATE kursi
        SET status_kursi = 'Tersedia'
        WHERE id_jadwal     = NEW.id_jadwal
```

```
          AND nomor_gerbong = NEW.nomor_gerbong
          AND nomor_kursi   = NEW.nomor_kursi;

    END IF;
END$$
DELIMITER ;
```

EVENTS

1. Pembatalan Tiket

```
DELIMITER $$

CREATE EVENT auto_cancel_tiket_30_menit
ON SCHEDULE EVERY 1 MINUTE
STARTS CURRENT_TIMESTAMP
DO
BEGIN
     UPDATE tiket
     SET status_pembayaran = 'Dibatalkan'
     WHERE status_pembayaran = 'Pending'
     AND waktu_pemesanan <= NOW() - INTERVAL 30 MINUTE;
END$$

DELIMITER ;
```

Tiket akan berstatus ‘Dibatalkan’ ketika status pembayaran ‘Pending’ dan tidak melakukan pembayaran setelah lebih dari 30 menit.